Math 110 — Week 2

Solutions

HW2 P1.

Idea. Sums can cancel, so look for the smallest cancellation.

False. Take $n = 1$, $v_1 \neq 0$ and $w_1 = -v_1$. Each list is independent (one nonzero vector), but $v_1 + w_1 = 0$ is a dependent list.

HW2 P2.

Idea. (a) every axiom holds in each coordinate. (b) a finite-dimensional space cannot contain arbitrarily long independent lists. (c) finite combinations of the e_j have only finitely many nonzero entries.

1. (a) $\mathbb{F}^\infty = \mathbb{F}^{\mathbb{N}}$ with coordinatewise operations. Each vector space axiom is an equation between sequences, and it holds in every coordinate because it holds in $\mathbb{F}$. Zero vector $(0, 0, \dots)$; $-(a_j) = (-a_j)$.
2. (b) For every m, $e_1, \dots, e_m$ is independent: $\sum \lambda_j e_j = (\lambda_1, \dots, \lambda_m, 0, \dots) = 0$ forces all $\lambda_j = 0$. If $\mathbb{F}^\infty$ were spanned by a list of length n, every independent list would have length $\leq n$ (Prop. 10). Take $m = n + 1$: contradiction.
3. (c) A finite combination $\lambda_1 e_{j_1} + \cdots + \lambda_k e_{j_k}$ is 0 outside the positions $j_1, \dots, j_k$. The sequence $(1, 1, 1, \dots)$ has no zero entries, so it is not such a combination.

HW2 P3.

Idea. Start from a spanning list that lies in $U \cup W$ and thin it out; deleting vectors keeps the rest in $U \cup W$.

1. U and W are finite-dimensional (Prop. 14). Let $u_1, \dots, u_k$ be a basis of U and $w_1, \dots, w_l$ a basis of W.
2. The list $u_1, \dots, u_k, w_1, \dots, w_l$ spans $U + W = V$.
3. By Prop. 9 some of its vectors can be deleted to leave a basis of V. Every remaining vector is one of the u_i or w_j, so it lies in $U \cup W$.

HW2 P4.

Idea. $\dim \mathbb{F}^2 = 2$, so a proper nonzero subspace can only have dimension 1, and two vectors form a basis exactly when they are independent.

1. (a) Let U be a subspace with $\{0\} \neq U \neq \mathbb{F}^2$. By Prop. 14, $\dim U \leq 2$, and $\dim U \neq 2$ since $U \neq \mathbb{F}^2$; $\dim U \neq 0$ since $U \neq \{0\}$. So $\dim U = 1$: U has a basis consisting of one vector $v \neq 0$, and $U = \operatorname{span}(v)$.
2. (b) ($\Rightarrow$) If $w = cv$, then $cv - w = 0$ is a nontrivial relation, so v, w is dependent; same if $v = cw$.
3. ($\Leftarrow$) Suppose neither is a multiple of the other, and $\lambda v + \mu w = 0$. If $\mu \neq 0$ then $w = -\frac{\lambda}{\mu} v$; if $\lambda \neq 0$ then $v = -\frac{\mu}{\lambda} w$. Both are excluded, so $\lambda = \mu = 0$. Two independent vectors in a 2-dimensional space form a basis (Prop. 13).

HW2 P5.

Idea. The dimension formula turns "$U + W$ is everything and the dimensions add up" into "the intersection is 0".

$\dim(U \cap W) = \dim U + \dim W - \dim(U + W) = 3 + 5 - 8 = 0$, so $U \cap W = \{0\}$ and the sum is direct.

HW2 P6.

Idea. Both conditions are about the point 1, so expand p in powers of $x-1$: then $p(1)$ and $p'(1)$ are the first two coefficients.

1. (a) The zero polynomial is in U. If $p, q \in U$: $(\lambda p + q)(1) = \lambda p(1) + q(1) = 0$ and $(\lambda p + q)'(1) = \lambda p'(1) + q'(1) = 0$.
2. (b) $1, (x-1), (x-1)^2, (x-1)^3, (x-1)^4$ is a basis of $\mathcal{P}_4(\mathbb{R})$ (Ex. 18). Write $p = c_0 + c_1(x-1) + c_2(x-1)^2 + c_3(x-1)^3 + c_4(x-1)^4$.
3. $p(1) = c_0$ and $p'(1) = c_1$. So $p \in U \iff c_0 = c_1 = 0 \iff p \in \operatorname{span}\big((x-1)^2, (x-1)^3, (x-1)^4\big)$.
4. These three polynomials are independent (part of a basis). So they are a basis of U and $\dim U = 3$.

R1.

Idea. Four vectors in $\mathbb{R}^4$: if they are independent they are a basis, and then both $U + W = \mathbb{R}^4$ and $U \cap W = \{0\}$ follow.

1. (a) Suppose $au_1 + bu_2 + cw_1 + dw_2 = 0$. Coordinates: $a + d = 0$, $a + b = 0$, $b + c = 0$, $c - d = 0$. So $d = -a$, $b = -a$, $c = a$, and the last equation gives $a - (-a) = 2a = 0$. Hence $a = b = c = d = 0$.
2. So u_1, u_2, w_1, w_2 is an independent list of $4 = \dim \mathbb{R}^4$ vectors, a basis (Prop. 13). Hence $U + W = \mathbb{R}^4$, and $\dim U = \dim W = 2$.
3. $\dim(U \cap W) = 2 + 2 - 4 = 0$, so the sum is direct.
4. (b) Solve $au_1 + bu_2 + cw_1 + dw_2 = (2, 1, 3, 0)$: $a + d = 2$, $a + b = 1$, $b + c = 3$, $c - d = 0$. From the last, $c = d$; then $b = 3 - d$, $a = 2 - d$, and $a + b = 5 - 2d = 1$ gives $d = 2$. So $a = 0$, $b = 1$, $c = d = 2$.
5. $u = u_2 = (0, 1, 1, 0)$ and $w = 2w_1 + 2w_2 = (2, 0, 2, 0)$. Check: $u + w = (2, 1, 3, 0)$.

R2.

Idea. Extend a basis of U to a basis of V; the added vectors span the complement.

1. Let $u_1, \ldots, u_k$ be a basis of U (Prop. 14) and extend it to a basis $u_1, \ldots, u_k, w_1, \ldots, w_l$ of V (Prop. 11). Let $W = \operatorname{span}(w_1, \ldots, w_l)$.
2. $U + W$ contains a basis of V, so $U + W = V$.
3. If $x \in U \cap W$, then $x = \sum a_i u_i = \sum b_j w_j$, so $\sum a_i u_i - \sum b_j w_j = 0$. The combined list is independent, so all a_i, b_j are 0 and $x = 0$.

R3.

Idea. Write the condition as one linear equation in the coefficients and solve it with free parameters.

1. For $p = a + bx + cx^2$: $p'(1) = b + 2c$ and $p(2) = a + 2b + 4c$. The condition is $a + b + 2c = 0$, i.e. $a = -b - 2c$.
2. So $p = b(x-1) + c(x^2 - 2)$ with b, c free: $V = \operatorname{span}(x - 1,\ x^2 - 2)$. Check: for $x - 1$, $p'(1) = 1 = p(2)$; for $x^2 - 2$, $p'(1) = 2 = p(2)$.
3. They are independent (different degrees). So $x - 1,\ x^2 - 2$ is a basis and $\dim V = 2$.